

Aurora Food Cold-Chain Validation

ACV-26-114 - experimental report with figures and footnotes

Objective

This study evaluates insulated food containers on the Aurora regional route under matched loading and weather conditions. The validation identifier is ACV-26-114. It does not cover medicine, biological samples or consumer refrigeration, and its findings must not be generalized beyond the defined route and packaging system.

Study design

Twenty-four monitored shipments were paired by departure window, payload class and forecast temperature. Twelve used the baseline liner and twelve used the revised liner. Data loggers recorded at two-minute intervals, while handling events were confirmed from depot scans and vehicle records.

Primary outcome

The primary metric is the time-weighted mean internal temperature during the transport window. Excursion counts and door-open duration are secondary outcomes. A packaging change is considered useful only when it improves temperature control without increasing loading errors.

Evidence limits

Driver comments explain unusual stops but do not replace logger data. The report contains no customer identities, delivery addresses, commercial prices or access credentials. Questions requesting those facts are unsupported.

1. Baseline shipments

First half of the matched comparison

Baseline result

Across the twelve baseline-liner shipments, the time-weighted mean internal temperature was 5.6 degrees C. The median shipment mean was 5.5 degrees C and the highest single-shipment mean was 6.4 degrees C. The matched comparison uses 5.6 degrees C.

Exposure context

Outdoor temperature ranged from 18.2 to 31.7 degrees C. Two routes included planned depot waits, and one included an unplanned traffic stop. These events remained in the primary analysis because equivalent handling categories were represented after the packaging change.

Logger quality

All twenty-four primary loggers passed calibration checks. Two backup loggers contained short gaps, but neither supplied a primary outcome. Clock alignment was corrected before pairing, and the original timestamps remain available in the audit record.

Nearby values

A warehouse energy dashboard reports 4.9 degrees C for a fixed cool room. That value has a different measurement boundary and is not the shipment baseline. A training slide uses 6.0 degrees C as an example specification.

2. Revised-liner shipments

Second half of the matched comparison

Observed result

Across the twelve revised-liner shipments, the time-weighted mean internal temperature was 4.8 degrees C. Door-open duration was similar to baseline, and loading-error checks did not increase. This result must be combined with the preceding page for a before-and-after answer.

Derived change

The validated means differ by 0.8 degrees C, equivalent to a 14.3 percent reduction relative to baseline. The report publishes one-decimal means and calculates the percentage from unrounded logger aggregates.

Robustness

Removing the longest traffic delay changed the reduction by less than 0.1 degrees C. Pair-level results favored the revised liner in ten of twelve pairs. The study remains too small to estimate rare seal failures.

Recommendation

Continue the revised liner for a three-month operational trial while preserving the same logger placement and pairing rules. Changing the payload mix or sensor location would create a new series.

3. Excursions by route

Native table with route-level evidence

Definition

An excursion is a continuous interval above 7.0 degrees C lasting at least ten minutes. Shorter observations remain in the raw series but do not count toward the table.

Route	Shipments	Baseline	Revised	Longest revised
Aurora North	6	4	1	12 min
Lake Junction	6	3	2	18 min
Pine Ridge	6	5	2	16 min
Coastal Link	6	2	0	0 min

Interpretation

Pine Ridge recorded five baseline excursions and two revised-liner excursions. Coastal Link recorded no revised-liner excursion, but its shorter route prevents a direct claim that packaging alone caused the result.

Handling review

The Lake Junction revised excursions coincided with an extended depot handover. The event was retained because the door scans and logger sequence were complete.

4. Sensor-position analysis

Raster-only chart with unique positions

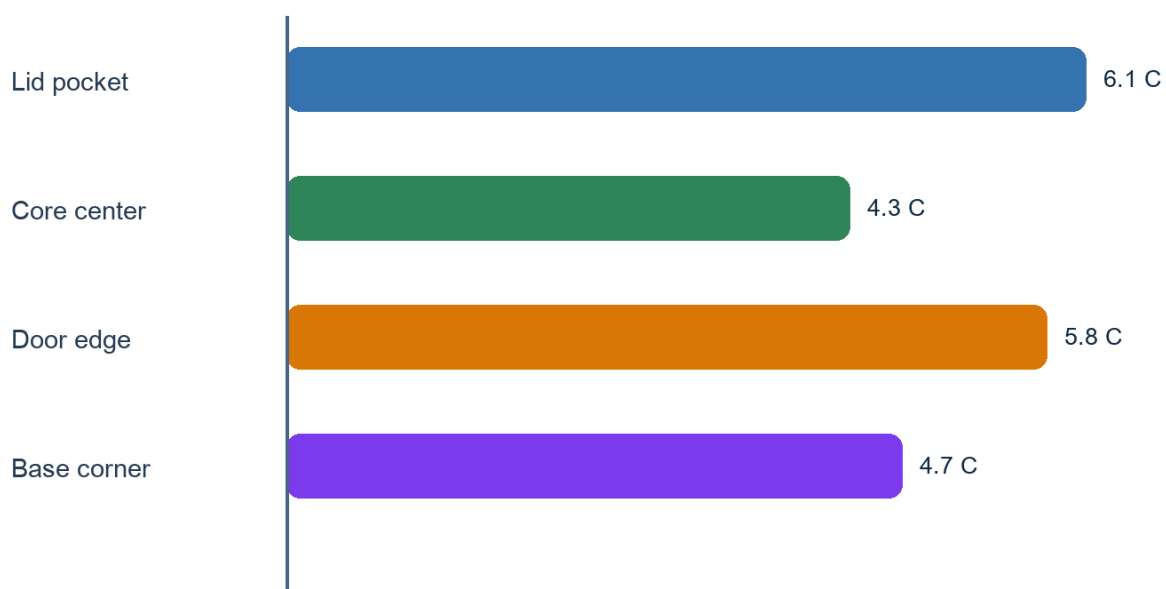
Figure scope

Figure 4 compares mean temperatures at four sensor positions used only in a supplemental test. The position names and values are not repeated in native text. Visual evaluation must read the plotted labels rather than infer a value from the primary outcome.

Interpretation boundary

The supplemental test used one instrumented load and is descriptive. It helps identify internal gradients but does not replace the twenty-four-shipment comparison.

Supplemental mean temperature by sensor position



5. Approval, limitations and negative evidence

Version status, rejected criteria and citation rules

Approval

Report version 1.0 was approved on 3 November 2026 by the Packaging Validation Lead. A September draft is not an approved source.

Rejected setpoint

The draft defined excursions above 6.5 degrees C for five minutes. Reviewers rejected that rule. The approved definition is above 7.0 degrees C for at least ten minutes.

Limitations

The sample supports route-specific comparison but not rare-failure rates, other foods or different container sizes.

Unsupported requests

No customer, price, delivery address, driver phone number or depot access code appears in this report.

Citation rule

Before-and-after answers cite pages 2 and 3; route counts cite page 4; sensor-position values cite the chart region on page 5.